A Complete Floquet–Jordan Atlas for the Two-Step Hill Oscillator

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Abstract

We classify the periodic two-step Hill oscillator for arbitrary real segment coefficients, including oscillatory, zero-frequency, and hyperbolic pieces. Entire segment functions give one exact $SL(2,\mathbb{R})$ monodromy and a closed Floquet discriminant whose elliptic and hyperbolic regions are complete.

1 One entire transfer for all segment signs

Fix the periodic equation

$$y'' + k(t)y = 0, \qquad k(t) = \begin{cases} k_1, & 0 \le t < \tau_1, \\ k_2, & \tau_1 \le t < T, \end{cases} \qquad T = \tau_1 + \tau_2 > 0, \tag{1}$$

with $k_j \in \mathbb{R}$ and $\tau_j \ge 0$. Define entire functions

$$C(k,t) = \sum_{m\ge 0} \frac{(-k)^m t^{2m}}{(2m)!}, \qquad S(k,t) = \sum_{m\ge 0} \frac{(-k)^m t^{2m+1}}{(2m+1)!}. \tag{2}$$

They obey $C_t = -kS$, $S_t = C$, and $C^2 + kS^2 = 1$. Hence the state $Y = (y, y')^\mathsf{T}$ advances through one constant segment by

$$\Phi(k,t) = \begin{pmatrix} C & S \\ -kS & C \end{pmatrix}, \qquad \det \Phi = 1. \tag{3}$$

For $k > 0$, $k = 0$, and $k < 0$, this is respectively the cosine–sine, shear, and cosh–sinh transfer, without a square-root branch convention.

2 Complete stability and boundary theorem

Put $M = \Phi(k_2, \tau_2)\Phi(k_1, \tau_1)$ and write C_j, S_j for the two segment values. Direct multiplication gives

$$\det M = 1, \qquad \Delta = \operatorname{tr} M = 2C_1C_2 - (k_1 + k_2)S_1S_2. \tag{4}$$

Theorem 1. *For the oscillator* (1)*:*

1. *If $|\Delta| < 2$, the monodromy is elliptic and every solution is bounded. Its Floquet angle per unit time is $T^{-1}\arccos(\Delta/2)$.*
2. *If $|\Delta| > 2$, the multipliers are a real reciprocal pair. Generic solutions grow exponentially, one forward eigenline decays, and the positive growth rate is $T^{-1}\operatorname{arcosh}(|\Delta|/2)$.*
3. *If $\Delta = \pm 2$ and $M = \pm I$, every solution is periodic or antiperiodic over one coefficient period. If instead $M \ne \pm I$, then $\operatorname{rank}(M \mp I) = 1$: one Floquet line is bounded and generic solutions grow linearly, with alternating sign on the minus face.*

For every $n \ge 1$,

$$M^n = U_{n-1}(\Delta/2)M - U_{n-2}(\Delta/2)I. \tag{5}$$

Here $U_{-1} = 0$ and $U_0 = 1$, so (5) *includes $n = 1$.*

Proof. The characteristic polynomial is $\lambda^2 - \Delta\lambda + 1$. Its roots are distinct unit conjugates for $|\Delta| < 2$ and distinct real reciprocals for $|\Delta| > 2$, proving the first two cases. At trace ± 2, a real 2×2 determinant-one matrix is either $\pm I$ or has one Jordan block; its powers distinguish bounded from linear growth. Cayley–Hamilton and the recurrence $U_n(x) = 2xU_{n-1}(x) - U_{n-2}(x)$ prove (5).

The baseline ends with the exact monodromy and stability theorem. Executable receipts and the degenerate-face ledger enter only in later revisions.

3 Conclusion and limitations

The all-sign entire transfer, one discriminant, the scalar/Jordan split, and the Chebyshev law together close the two-step family. This does not classify arbitrary periodic coefficients, nonlinear perturbations, or every initial condition's forward behavior in the hyperbolic case.

References

[1] G. W. Hill, "On the part of the motion of the lunar perigee which is a function of the mean motions of the sun and moon," *Acta Mathematica* 8 (1886), 1–36, doi:10.1007/BF02417081.

[2] Y. F. Golubev, "Resonances of linear differential equations with piecewise constant coefficients," Keldysh Institute Preprint 43 (1997), MathNet record.